GEOG 172

Scale and the Openshaw Effect

Learn about the modifiable areal unit problem through the Openshaw Experiment.

Lecture 5-1

RESEARCHER DECISIONS

- Statistical procedures are math. They are algorithms that process inputs into outputs.
- Statistical arguments are arguments. The depend on the reasoning behind the selection in input, the use of methods, the interpretation of outputs.
- Spatial statistics have lots of inputs Changing them will change the outputs, and may change their interpretation.



The MAUP FAMILY FEUD

To check if you did the readings!

ACTIVITY

Game Rules:

- Pick your family: **The McMappers vs. The O'Boundaries!**
- Start with 10 points per family.
- Send up one player from each family, new question, new players!
- **Read the statement:** TRUE or FALSE?
- **Buzz when ready!** Both players lock in their answers before either reveals. First buzzer reveals first.
- **Correct?** Gain the question's points. Incorrect? Lose that many.
- Both families score independently: both can gain or lose points.
- **Highest final score wins!**



"We asked 99 counties in Iowa ……


1

2 POINTS

The Modifiable Areal Unit Problem is a single problem caused only by changing the size of spatial units.

FALSE


1

2 POINTS

The scale problem occurs when the same data are combined into progressively larger areal units.

TRUE


1

3 POINTS

The aggregation problem can occur even when the number of spatial units stays the same.

TRUE


1

2 POINTS

In Openshaw's terminology, grouping systems require contiguous areas, while zoning systems do not.

FALSE


1

1 POINT

In the Iowa example, the original observations are the 99 counties in Iowa.

TRUE


1

1 POINT

The two variables in the Iowa example are Republican voting and the percentage of the population aged over 65.

TRUE


1

1 POINT

At the original 99-county level, the correlation between the two Iowa variables is approximately +0.35.

TRUE


1

3 POINTS

Once the two variables are fixed, there is only one statistically valid way to aggregate the 99 Iowa counties into six regions.

FALSE


1

3 POINTS

Different six-region arrangements of exactly the same Iowa counties can produce different correlation coefficients.

TRUE


1

2 POINTS

Openshaw used an automated procedure to search for aggregations producing very high and very low correlations.

TRUE


1

2 POINTS

As the number of zones increases toward the original 99 counties, the range of possible correlation coefficients generally becomes narrower.

TRUE


1

2 POINTS

Openshaw generated only one alternative zoning at each spatial scale because one zoning is enough to measure the MAUP.

FALSE


1

3 POINTS

Grouping and zoning produce identical distributions of correlation coefficients as long as they contain the same number of units.

FALSE


1

3 POINTS

The interaction between spatial autocorrelation and the contiguity constraint helps explain why zoning behaves differently from unconstrained grouping.

TRUE


1

2 POINTS

Openshaw concludes that we can use a universal rule to predict exactly how much a correlation will change whenever data are aggregated.

FALSE

Understanding Openshaw's Experiment

Examining a classic

SUMMARY

QUESTION How much does the relationship between age composition and Republican voting depend on how counties are combined?

DATA 99 Iowa counties, with data on the percentage of older residents and the percentage of Republican votes.

APPROACH Keep the original county data fixed. Combine counties in different ways and recalculate the correlation.

TWO PROBLEMS IN THE EXPERIMENT

SCALE	Variation in results observed when the same data are combined into sets of increasingly larger units
AGGREGATION	Variation in results where the number of units *n* is held constant

KEY DISTINCTION

How many units? versus Which counties belong together?

AIGORITHMS IN OPENSHAW'S FIRST EXPERIMENT

ZONING Combining counties into regions with a contiguity requirement. Counties within each region must form one connected area.

GROUPING Combining counties into groups without a contiguity requirement. Counties in the same group may be geographically separated.

KEY DISTINCTION

Zoning is grouping with a spatial constraint.

AN EXAMPLE OF ONE ITERATION

DEFINE	Choose the number of regions or groups and whether contiguity is required.
COMBINE	Assign each county to a region or group using the selected algorithm (grouping or zoning).
SUMMARIZE	Aggregate the county data to produce one pair of variable values for each region or group.
CALCULATE	Compute the correlation across those new observations.

REPEAT + RECALCULATE 10,0000 TIMES

OPENSHAW'S FIRST EXPERIMENT

Table 5.2. Maximum and minimum values of the correlation coefficient.

Number of zones or groups	Zoning systems [a]		Grouping without contiguity	
	minimum r	maximum r	minimum r	maximum r
6	−0·999	0·999	−0·999	0·999
12	−0·984	0·999	−0·999	0·999
18	−0·936	0·996	−0·977	0·999
24	−0·811	0·979	−0·994	0·999
30	−0·770	0·968	−0·989	0·999
36	−0·745	0·949	−0·987	0·998
42	−0·613	0·891	−0·980	0·996
48	−0·548	0·886	−0·967	0·995
54	−0·405	0·823	−0·892	0·983
60	−0·379	0·777	−0·787	0·983
66	−0·180	0·709	−0·698	0·953
72	−0·059	0·703	−0·579	0·927

[a] Best from fifteen different random zoning systems used as starting points for the automatic zoning algorithm.

Reconstructing Openshaw's Workflow

ACTIVITY

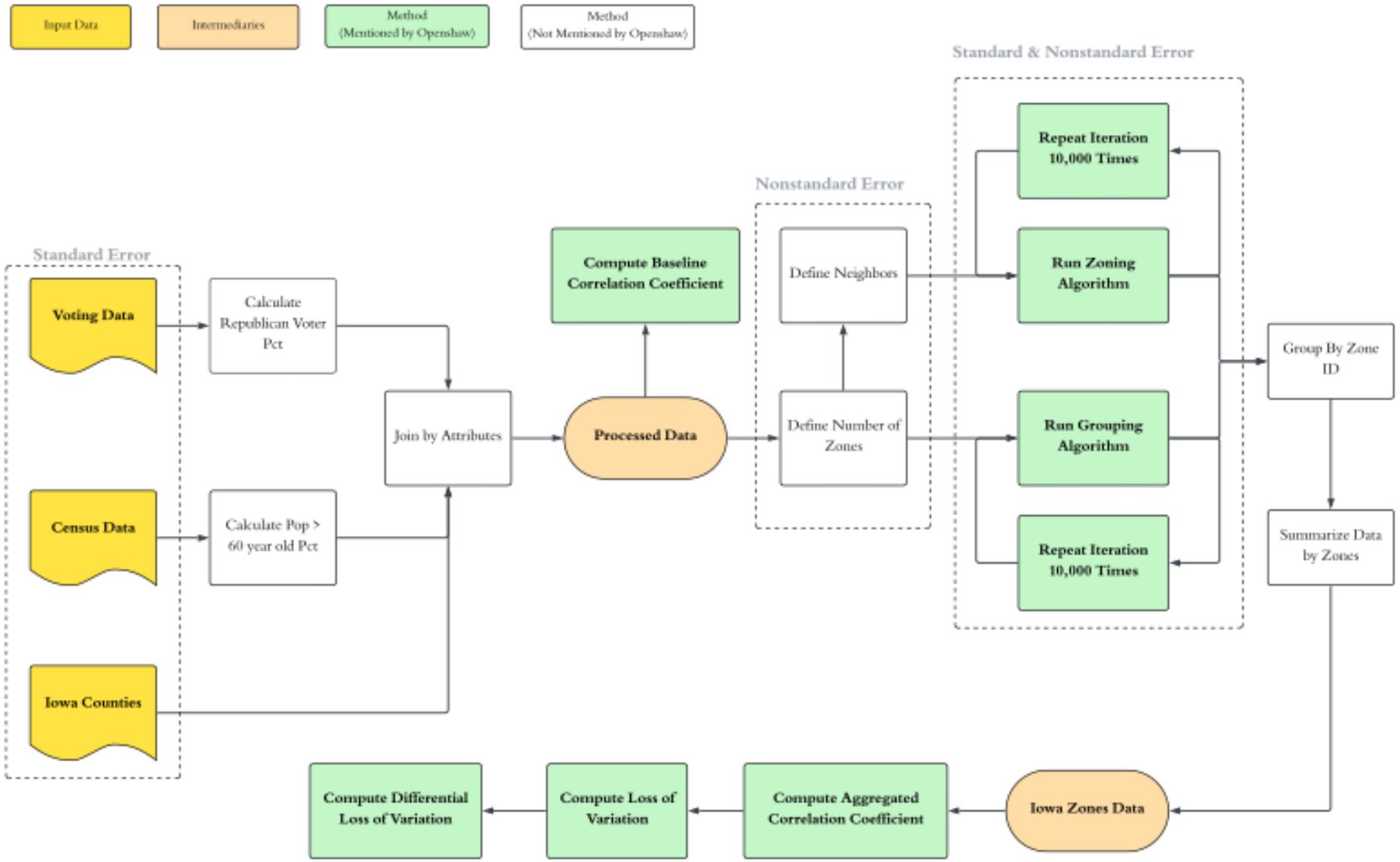
Purpose: To assess your understanding of the original Openshaw experiment and practice research design and planning

Task:

- **Form groups of 4–5** and collect your workflow cards, paper, and markers.
- **Sort your cards by color:** yellow = input data; orange = intermediate datasets; green = methods mentioned by Openshaw; white = methods not mentioned.
- **Start with the three inputs:** voting data, census data, and Iowa counties. Arrange the steps needed to prepare and combine them.
- **Build the analysis and draw arrows to connect the cards.** Show which steps feed into others and where the 10,000 iterations loop back.

DEBRIEF

- Compare your workflow to others. What differences do you observe? Are these differences about ordering, missing steps, or interpretation?
- Many steps in this workflow involve researcher decisions. Which decisions are explicitly specified by Openshaw? Which are left implicit or underspecified?
- Did you identify any potential issues or limitations in Openshaw's original experiment?
- **Consider the role of scale: Where in the workflow is scale being fixed, changed, or assumed?**
- What assumptions are we making about Iowa by aggregating it in this way?


Input Data
Intermediaries
Method (Mentioned by Openshaw)
Method (Not Mentioned by Openshaw)
Standard & Nonstandard Error
Nonstandard Error
Standard Error
Repeat Iteration 10,000 Times
Run Zoning Algorithm
Run Grouping Algorithm
Repeat Iteration 10,000 Times
Compute Baseline Correlation Coefficient
Define Neighbors
Define Number of Zones
Voting Data
Calculate Republican Voter Pct
Census Data
Calculate Pop > 60 year old Pct
Iowa Counties
Join by Attributes
Processed Data
Group By Zone ID
Summarize Data by Zones
Iowa Zones Data
Compute Aggregated Correlation Coefficient
Compute Loss of Variation
Compute Differential Loss of Variation